REFLECTION PAPER

PROVIDERS OF DECENTRALIZED BLUE-GREEN FLOOD ADAPTATION MEASURES PROTECT OTHERS, NOT THEMSELVES, EVEN IN FLAT URBAN TERRAIN

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Summary

This article looks at pluvial flooding, the kind caused by heavy rainfall rather than overflowing rivers, and asks how cities can manage it through small-scale blue-green infrastructure: rain gardens, permeable pavement, retention ponds built to serve a single household or a neighborhood block. Roggero, Brill, and Tügel ran hydrodynamic simulations of extreme rainfall across Berlin, a city known for its flat terrain, to test a widely held policy assumption: does building your own stormwater system actually make your household the safest from flooding?

The answer is no. Even on flat ground, upstream and downstream relationships between neighboring plots still exist at a micro scale, and they don't disappear just because the slope is gentle. As a result, the household that pays for the infrastructure often isn't the one that benefits most; that benefit tends to flow downstream to a neighbor instead. This is a collective-action problem: once the payoff might land on someone else's property, the incentive to invest on your own dime shrinks, and people wait around for a neighbor to act first. The authors conclude that voluntary individual effort won't be enough here. What's needed instead are targeted subsidies, collective obligations set at the block level, or planning schemes based on how water actually flows rather than broad assumptions about a city's topography.

Reflection

What struck me most is how directly this challenges an assumption baked into much green-infrastructure policy: that whoever pays for the system automatically protects it. The hydrodynamic modeling here is detailed enough to show that runoff keeps moving in fairly predictable directions and patterns even when the slope is barely noticeable, and it's that flow pattern, not who funded the infrastructure, that ultimately decides who stays dry. Combining hydrology with institutional economics this way lets the findings speak to several audiences at once, including urban engineers, policymakers, and environmental economists.

A limitation worth noting is that the study only examines one temperate city where rainfall is spread fairly evenly through the year, so applying its conclusions directly to tropical cities with sudden, intense convective storms should be done carefully. That said, the underlying logic still holds up outside this context: the costs and benefits of flood infrastructure can fall in different places geographically, regardless of a city's climate or terrain.

Relevance to Indonesia

A similar pattern likely applies to flat, flood-prone cities in Indonesia, including Jakarta, Semarang, Pekalongan, and parts of Surabaya. A large part of Indonesia's flood-mitigation policy leans on decentralized micro-infrastructure like infiltration wells, biopori holes, and household retention ponds, either required under building-permit regulations (PBG) or promoted through sponge-city programs. These policies rest on the same assumption: that a household installing this kind of infrastructure is making itself safer.

Roggero, Brill, and Tügel's findings suggest the same reversal could happen here too: small-scale topographic flow can send the benefit downstream to a neighbor instead of back to whoever paid for the infrastructure. And when people cover the cost alone while the payoff lands elsewhere, their motivation to keep participating tends to drop off. What local governments would need is hydrological mapping done at the micro level, down to the neighborhood or block rather than just the broader watershed, combined with incentive structures and collective-obligation schemes at the RT/RW level that actually reflect who benefits. Getting this right could distribute the costs and benefits of climate adaptation more fairly and help blue-green infrastructure spread more evenly instead of appearing only in scattered pockets.

Source: *Roggero, M., Brill, F., & Tügel, F. (2025). Providers of decentralized blue-green flood adaptation measures protect others, not themselves, even in flat urban terrain. International Journal of Disaster Risk Reduction, 128, 105721.* https://doi.org/10.1016/j.ijdrr.2025.105721